Sam Dhole, Ph.D.

Materials Scientist — Radiation Detection — Chemical Regulatory Compliance — Lab Automation

samdhole95@gmail.com — +1 (716) 603-1384 — Google Scholar (300+ citations)
Remote Consulting — US & International

Professional Summary

Ph.D. Materials Scientist with 8+ years of R&D experience spanning academic research and industrial product development. Led multiple DOE SBIR projects from Phase I through Phase II at CapeSym Inc., delivering prototype radiation detectors for medical imaging, nuclear physics, and homeland security applications. Deep expertise in perovskite and quantum dot materials, thin film deposition, optical/electrical characterization, and scientific automation (Python, LabVIEW, MATLAB). Dual undergraduate degrees in Chemistry and Electronics & Instrumentation provide cross-domain fluency from wet-bench synthesis to hardware integration to data science.

Core competencies: DOE SBIR project execution — Materials characterization (XRD, SEM, FTIR, Raman, PL, UV-Vis) — Thin film fabrication (PVD, solution processing) — Lab automation & data pipelines — Chemical regulatory compliance (TSCA, CEPA, GHS/SDS) — Technical documentation & proposal writing

Professional Experience

Haloforge Labs — Independent Consultant **Remote — Nov 2025–present**

Advanced materials and specialty chemical consulting for R&D teams, startups, and investors.

- GHS/HazCom technical review: SDS authoring and hazard classification for advanced materials (perovskites, quantum dots, specialty chemicals, nanomaterials)
- SBIR/STTR technical red-team: proposal critique, milestone feasibility review, and progress report drafting for DOE and other federal programs
- Materials characterization interpretation: XRD, SEM/EDX, spectroscopy data analysis and client-ready reporting
- Technical documentation: risk memos, literature reviews, and experimental plans for deeptech and early-stage materials companies

CapeSym, Inc. — Scientist **Natick, MA — 2022–2025**

Small R&D company (11–50 employees) developing novel materials, sensors, and instruments for radiation detection. PI/Supervisor: Amlan Datta, Senior Scientist & R&D Principal Investigator.

DOE SBIR Project 1: Low-Cost Pixelated X-Ray Imager

Phase I + II — Contract DE-SC0022472 — 2022–2025

- Lead scientist on MAPI ($MAPbI_3$) thick-film direct-conversion X-ray detector development
- Fabricated detectors on two a-Si:H active pixel array sizes: 3.5×3.5 cm (150 μm pixels) and 10×10 cm
- Achieved X-ray sensitivity of 1.6×10^5 μC Gy^{-1} A^{-1} — highest $S/\sqrt{J}$ ratio in published MAPI literature at time
- Dark current: 0.5 nA/cm^2 at 50V bias; stable 60 days in ambient atmosphere without encapsulation
- Developed 3 novel fabrication processes: (1) methylamine gas recrystallization, (2) TMPTA polymer crosslinking + soft-pressing, (3) MAPI-polyimide composite hole blocking layer
- **Outcome**: Clinched Phase II funding; technology advancing toward medical X-ray imaging prototype

DOE SBIR Project 2: High Spatial Resolution Detectors for Nuclear Physics

Phase I + II — Contract DE-SC0023561 — 2022–2026

- Developed MP-OGS (Microcapillary Plate + Organic Glass Scintillator) particle tracking detectors
- Phase I: Fabricated crack-free MP-OGS in 100 μm and 20 μm pore sizes; achieved 2.45 ns decay time, PSD FoM = 2.8 (gamma/neutron separation)
- Designed CapeSym SiPM readout PCB (128-channel S13552 array, 230 μm pitch) with custom 9V low-ripple power supply
- Phase II: Pivoted to NP-OGS nanoguide format; achieved **5 lp/mm spatial resolution** — exceeding 2× the theoretical 230 μm SiPM limit
- Built advanced qCMOS imaging system (Hamamatsu Orca Quest): tracked individual electrons (28 μm trajectory, 0–4 photon counting)
- Designed complete optical system in OpenSCAD: lens adapters, sample holders, camera mounts, mirror holders for beamline deployment
- Scale-up: demonstrated 100×100 mm NP-Plastic commercial scalability
- Collaborators: Sandia National Labs, Fermilab, University of Tennessee, FRIB, LANSCE

DOE SBIR Project 3: High QE Scintillators for Dual Readout Calorimetry

Phase I — Contract DE-SC0024780 — Feb–Nov 2024

- Synthesized CZSM (Mn-doped $Cd_{0.2}Zn_{0.8}S$/ZnS core-shell) quantum dots via hot injection Schlenk line
- Achieved **250 nm Stokes shift** (excitation 340 nm → emission 590 nm) — tunable via Cd:Zn ratio
- Created QD-PMMA composites at up to 25 wt% loading with maintained optical transparency
- Demonstrated simultaneous Cerenkov + scintillation detection with temporal and wavelength separation
- Screened 4 QD systems; developed surface modification protocols for polymer matrix compatibility
- Collaborator: James Freeman, Fermilab

DHS-CWMD Project: Engineered Scintillator Ceramics

Contracts 70RWMD23C0000017/18 — Jul 2023–2025

- Developed Cs_2HfCl_6 (CHC) ceramic scintillator powder for dual gamma/neutron detection
- All CHC powder synthesis at CapeSym: dry, wet (toluene), and cryogenic (liquid N_2) ball milling
- Particle size control: 9–130 μm (various sieve meshes); multiple batches shipped to UCF for sintering
- Li:CHC (neutron-sensitized) and SrI_2 powders also processed
- Characterization by LBNL (Dr. Federico Moretti)

Predecessor Project: Synchrotron X-ray Detectors (DE-SC0019658)

Phase II Report #6 co-author — Jul–Dec 2022

- Executed full MAPI synthesis + fabrication pipeline: ACN/2-ME/GBL solvent system, automated blade coating
- Developed PI-MAPI composite hole blocking layer + MA gas treatment system (custom Ar/MA rotameter glovebox)
- Achieved record $S/\sqrt{J} = 2.07 \times 10^8$ μC Gy^{-1} cm^{-1} A$^{-1/2}$ — highest published at time
- >95% deposition uniformity over large area; 90-day ambient stability

Other CapeSym Contributions

- Parylene encapsulation (2 μm Parylene-C, Labcoter system); metallic contact deposition (Ag, Au, ITO)
- Custom argon glovebox protocols (<2 ppm O_2, <1 ppm H_2O) for moisture-sensitive perovskites
- Co-authored and sole-authored DOE quarterly progress reports across all projects
- Coordinated with external partners (Hamamatsu, SNL, Fermilab) for testing and component qualification

Los Alamos National Laboratory — Graduate Student Guest — CINT — 2021–2022

Host: Dr. Wanyi Nie, Center for Integrated Nanotechnologies (CINT).

- **First slot-die deposition of $MAPbI_3$ thick films**: Selected 2-methoxyethanol (bp 124°C, donor number 20 kcal/mol) to avoid solvent trapping endemic to DMF/DMSO in thick films; layer-by-layer scheme (prenucleation seed + thick layer) yielded 80–130 μm films with 59% large crystal domains
- Generalized process to lead-free $MA_3Bi_2I_9$ (bismuth halide): optimized single-layer 40 μm and two-layer 70 μm compact films; resistivity 2.31×10^6 Ω-cm
- Built portable environmental chamber for in-situ electrical measurements under humid CO_2
- Characterized films via SEM, XRD, Raman, PL, UV-Vis; Rutherford backscattering spectrometry (RBS, 2.3 MeV $^4He^+$) for quantitative stoichiometry at LANL NEC Pelletron
- Mentored undergraduate intern on photovoltaic device measurement

University at Buffalo — Graduate Research Assistant — Buffalo, NY — 2019–2022

Advisor: Prof. Quanxi Jia. GPA 3.967/4.000.

- Operated and managed High-Resolution XRD user facility (X'Pert PANalytical MRD PRO, Rigaku Ultima III); trained users and maintained equipment
- Established wet chemistry laboratory from scratch (purchasing, chemical inventory, safety protocols, OSHA compliance)
- **Polymer Assisted Deposition (PAD)**: Pioneered first aqueous chemical solution route for chalcogenide perovskite $BaZrS_3$ — 83 nm films on c-plane sapphire; Ba:Zr:S = 1.00:0.93:3.02 (RBS); PL peak 1.81 eV; Efros-Shklovskii variable range hopping transport confirmed; extended to $Ba_3Zr_2S_7$ (Ruddleson-Popper 2D) and $SrZrS_3$ on $LaAlO_3$ → *Photonics* 2023 publication
- **Resistive Switching**: Elucidated Schottky emission mechanism in $Na_{0.5}Bi_{0.5}TiO_3$ (NBT) memristors: mobile oxygen vacancy dynamics control ON/OFF ratio (∼11,000 at 600°C); designed MIM test structures (photolithography + PVD); also studied HfO_2:CeO_2 charge transport → *ACS Appl. Electron. Mater.* (2021) and *J. Mater. Chem. C* (2021)
- PPMS (Quantum Design) 4-probe R-T measurements on superconducting NbN films; ICP-OES for precursor stoichiometry verification → *Materials* 2023 publication
- **Thesis**: Synthesizing and Exploiting Perovskite Films for Desired Physical Properties: Oxides, Halides, and Chalcogenides (2022)

IIT Bombay — Research Intern **Mumbai, India — 2017–2018**

- Developed single-crystal growth protocols for halide perovskites; explored lead-free alternatives for PV
- Built and programmed automated Solar Cell I-V measurement system in LabVIEW

BITS Pilani — Student Researcher **Goa, India — 2015–2017**

- Optimized hydrothermal synthesis for $BiFeO_3$ nanowire-rGO nanocomposite
- DFT simulations (Quantum ESPRESSO) to elucidate interfacial polarization mechanisms
- Sol-gel synthesis of $BaTiO_3$ nanocrystals using EDTA complexation
- **Publication**: *J. Phys. Chem. C* (2017), >100 citations

Education

Ph.D., Materials Design and Innovation — University at Buffalo (SUNY) 2022
GPA: 3.967/4.000 — Thesis: Perovskite Films for Desired Physical Properties

M.S., Materials Design and Innovation — University at Buffalo (SUNY) 2020
GPA: 3.967/4.000 — Coursework: Statistics, Computational Chemistry, DOE, Nanotechnology

M.Sc.(Hons.) Chemistry + B.E.(Hons.) Electronics & Instrumentation — BITS Pilani 2018
CGPA: 7.45/10 First Division — Dual degree: chemistry + instrumentation engineering

Technical Skills

Semiconductor Processing: PVD (sputtering, evaporation), slot-die coating, spin-coating, blade coating, photolithography, wet/dry etching, plasma processing (ICP-RIE), cleanroom operations, glovebox (<2 ppm O_2/H_2O)

Materials Characterization: HR-XRD (epitaxial films, rocking curves, RSM), SEM/EDX, AFM, FTIR, Raman, UV-Vis, PL/TRPL, spectrofluorometry, CV-IV, impedance spectroscopy, Hall effect, PPMS (4-probe R-T), RBS (Rutherford backscattering), TGA/DSC, ICP-OES

Radiation Detection: SiPM readout systems, qCMOS cameras (Hamamatsu Orca Quest), PMT timing, scintillator characterization, X-ray imaging, neutron detection, PSD (pulse shape discrimination)

Data & Automation: Python (NumPy, SciPy, Pandas, Matplotlib, Langchain), MATLAB, LabVIEW (instrument control, test automation), JMP, R, DOE/statistical analysis, DFT (Quantum ESPRESSO)

Materials Synthesis: Perovskites (halide, oxide, chalcogenide), quantum dots (hot injection, ball milling), scintillators, functional oxides, polymer-matrix composites, Schlenk line techniques

Hardware Design: OpenSCAD (parametric CAD), 3D printing, optical system design, PCB design (KiCAD), vacuum system operation

Regulatory: TSCA, CEPA, GHS/SDS, hazardous materials handling, OSHA compliance, lab safety protocols

Publications (16 total, 300+ citations)

Selected First-Author Works:

- *Photonics* (2023): Facile Aqueous Solution Route for Chalcogenide Perovskite $BaZrS_3$ Films
- *Nanomaterials* (2022): Strain Engineering: Tunable Functionalities of Perovskite Metal Oxide Films (Review)
- *IEEE EDTM* (2021): Growth of Ferroic Metal Oxide Films by Polymer-assisted Deposition
- *J. Phys. Chem. C* (2017): $BiFeO_3$ Nanowire-RGO Nanocomposite (>100 citations)

Selected Co-Author Works:

- *Materials* (2023): CMOS-Compatible Superconducting NbN Films on 300 mm Si Wafer
- *Advanced Science* (2022): High Entropy Perovskite Thin Films
- *ACS Appl. Electron. Mater.* (2021): Electroforming-free HfO_2:CeO_2 Memristors
- *Nano Energy* (2021): Chalcogenide Perovskite $BaZrS_3$ Thin-Film Devices
- *IEEE NSS/MIC/RTSD* (2023, 2025 ×2): Radiation detector papers (CapeSym)

Freelance Service Offerings

Materials R&D Consulting

- Process development review
- Characterization interpretation
- Failure analysis
- DOE/SBIR proposal support

Lab Automation & Data

- LabVIEW instrument control
- Python data pipelines
- Automated test scripts
- Spectroscopy data processing

Regulatory Compliance

- GHS/SDS authoring
- TSCA new chemical notifications
- CEPA substance notifications
- HazCom 2024 compliance

Technical Writing

- SBIR proposal editing
- Experimental plan review
- Quarterly progress reports
- Technical memos